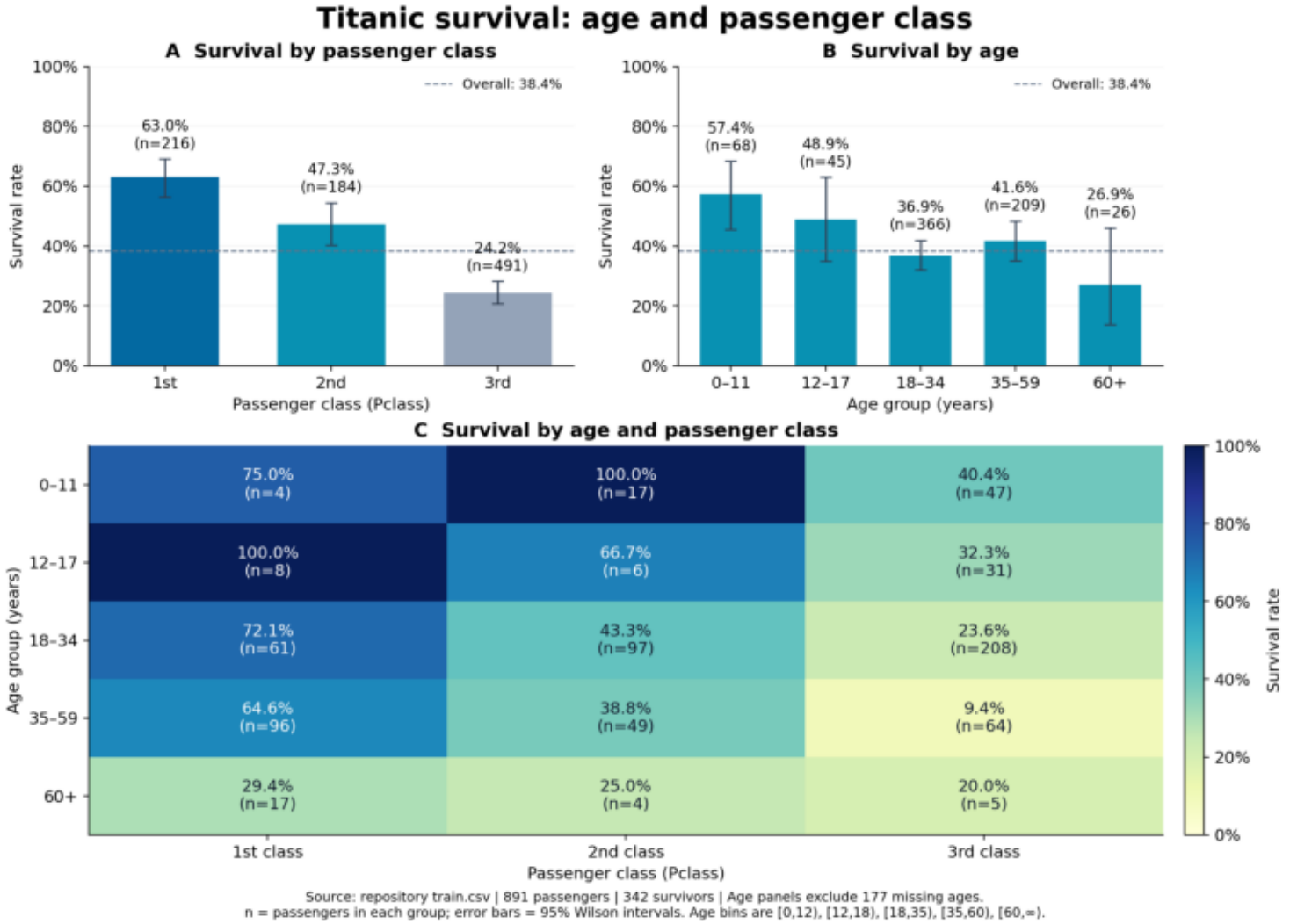
Titanic survival | Exploratory report

891 passengers • 342 survivors • Overall survival 38.4%



Key findings

• Class shows a strong survival gradient: 1st 63.0%, 2nd 47.3%, 3rd 24.2%.
The first-to-third class gap is 38.7 percentage points.

• Children aged 0–11 had 57.4% survival (n=68), versus 36.9%
for ages 18–34. Ages 60+ had 26.9% survival (n=26). The age pattern is not monotonic.

• Among children aged 0–11, survival was 100% in 2nd class (n=17),
but 40.4% in 3rd class (n=47). Age and class should be examined together.

Interpretation and limits
Age is missing for 177/891 passengers (19.9%); no ages were imputed.
Class totals use all passengers; age panels use the 714 recorded ages.
Small cells (especially older ages) are uncertain; 100% is an observed sample rate.
These are descriptive associations, not causal effects. Sex and other factors
may confound the patterns. Only train.csv has survival labels; test.csv is excluded.

Method: grouped observed survival proportions; 95% Wilson intervals on bars.